
Bachelor's Thesis Proposal

Well-formedness for the ν ActionGUI Modeling Languages

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1 Motivation

Model-driven privacy [1] (MDP) is a software development methodology in which privacy requirements are specified as models, and enforcement mechanisms are generated automatically via model transformations. MDP enables developers to specify three interconnected declarative models: a data model describing the state space of the system, a security model capturing fine-grained access control requirements (building on SecureUML [2]), and a privacy model expressing data protection requirements. From these models, the methodology automatically generates a web application together with enforcement mechanisms that make it security- and privacy-aware by design. The approach is implemented in a tool called ν ActionGUI [3], in which the three models are written as plain text files and fed to a generator.

Each modeling language has a *grammar* that defines which texts are syntactically valid. But being grammatically correct is not enough: a model must also satisfy a number of *well-formedness constraints* before it denotes a meaningful system. For example,

- for any data model, two entities must not share the same name, and an attribute name must be unique within its entity;
- given a data model, for any security model, an authorization rule in the security model must not reference an entity or attribute that does not exist in the data model; and
- given a data model and security model, for any privacy model, a personal data declaration in the privacy model must be a subset of the data model resource.

Such rules are typically captured as well-formedness constraints over the abstract syntax. For the ν ActionGUI modeling languages, some of these constraints are currently enforced in the tool but are not documented systematically. As a result, many violations are not caught. A precise, complete catalog of well-formedness constraints and informative violation reporting would make the approach more robust.

2 Objective

The objective of this thesis is to systematically identify, classify, and specify constraints for the three data, security, and privacy modeling languages in ν ActionGUI and the dependencies between them.

The constraints are then realized as a *validator* that takes a set of ν *ActionGUI* models and reports all violations, with their locations and messages, before the generator runs.

For each constraint, the thesis records (i) the model(s) to which it applies, (ii) a precise statement of the condition that a well-formed model must satisfy, (iii) its severity level (e.g., info, error, or warning), and (iv) an informative error message.

3 Tasks

1. Study the existing MDP approach and the textual modeling languages of ν *ActionGUI*.
2. Systematically collect the well-formedness constraints of the three models, classify them, and specify each precisely.
3. Implement a validator that checks a set of models against the constraints and reports violations with their locations and actionable messages.
4. Evaluate the catalog and the validator on existing case study models.
5. Document the methodology, the constraint catalog, and the evaluation in the final report.

4 Deliverables

The following deliverables are due at the end of the project:

Final report Please have a look at the submission requirements in the following section. The final report should consist of an introduction, a (theoretical) background section, sections describing the work packages and their outcomes, and a conclusion. The report must be written in English.

Constraint catalog and source code The catalog of well-formedness constraints, each with its scope, precise statement, severity, and error message, together with the complete, documented source code of the validator and a reproducible evaluation.

Presentation At the end of the thesis, a 20-minute presentation must be given in a thesis defense seminar. It should give an overview and discuss the most important highlights of the work.

References

- [1] Srdan Krstic, Hoang Nguyen, and David Basin. Model-driven privacy. *Proc. Priv. Enhancing Technol.*, 2024(1):314–329, 2024.
- [2] Torsten Lodderstedt, David Basin, and Jürgen Doser. SecureUML: A UML-based modeling language for model-driven security. In *UML 2002 — The Unified Modeling Language*, volume 2460 of *Lecture Notes in Computer Science*, pages 426–441. Springer, 2002.
- [3] Hoang Nguyen, Srdan Krstic, and David Basin. ν *ActionGUI*: Model-driven security and privacy, 2024.